TELECOM ParisTech

SD-TSIA205

Exercise 4

Splines function

Consider a pair of random variables $(X, Y) \in \mathbb{R}^d \times \mathbb{R}$. The regression function of Y on X is given by :

$$f^\star(x) = \mathbb{E}[Y|X = x]$$

The goal in nonparametric regression is to construct an estimation $\widehat{f}_N$ of $f^\star$ using i.i.d. samples $(x_1, y_1), \dots (x_N, y_N) \in \mathbb{R}^d \times \mathbb{R}$ with no particular assumptions on $\widehat{f}_N$ nor on the distribution of the data except that f is twice differentiable over some segment $[a, b]$ and that we can write :

$$y_n = f(x_n) + \varepsilon_n \ ,$$

with mean zero random errors $\varepsilon_1, \dots, \varepsilon_N$.
Minimizing the data fitting term $\sum_{n=1}^N |y_n - f(x_n)|^2$ over the span of $\mathcal{C}^2$ functions leads to *overfitting* : $\widehat{f}_N$ would be too complex and learn the noise ε. To avoid learning functions that are too "wiggly", a good tradeoff is to minimize instead the regularized problem :

$$\min_{f \in \mathcal{C}^2} \sum_{n=1}^N |y_n - f(x_n)|^2 + \lambda \int_a^b f''(x)^2 dx \ , \tag{1}$$

for some fixed hyperparameter $\lambda > 0$.

A practical characterization of the splines solution : We know that the solution of Problem (1) is given by splines function and so enable us to go back to a finite dimensional problem since all we need to do is find the coefficients of the cubic polynomials that define it. Let g be the so-called splines at points $x_1, \dots, x_N$ and define the quantities

$$g_n = g(x_n) \text{ and } \gamma_n = g''(x_n).$$

Notice that since g is linear outside $[x_1, x_N]$ (i.e on $[a, x_1]$ and $[x_N, b]$), it follows that $\gamma_1 = \gamma_N = 0$. Now define $\mathbf{g} = (g_1, \dots, g_N)^\top \in \mathbb{R}^N$ and $\boldsymbol{\gamma} = (\gamma_2, \dots, \gamma_{N-1}) \in \mathbb{R}^{N-2}$. Finally, let $h_n = x_{n+1} - x_n$ for $n = 1 \dots N-1$ and define the symmetric matrix $\mathbf{R} \in \mathbb{R}^{N-2, N-2}$ and the rectangular matrix $\mathbf{Q} \in \mathbb{R}^{N, N-2}$ with column indexing $j = 2 \dots N-1$ in order to fit the indexing of γ (the top left element of $\mathbf{Q}$ is thus $\mathbf{Q}_{12}$) such that :

$$\begin{cases} \mathbf{Q}_{j-1,j} = \frac{1}{h_{j-1}} \\ \mathbf{Q}_{jj} = -\frac{1}{h_{j-1}} - \frac{1}{h_j} \\ \mathbf{Q}_{j+1,j} = \frac{1}{h_j} \\ \mathbf{Q}_{ij} = 0 \text{ if } |i-j| \geq 2 \end{cases} \tag{2}$$

$$\begin{cases} \mathbf{R}_{ii} = \frac{1}{3}(h_{i-1} + h_i) \text{ for } i = 2, \dots, n-1 \\ \mathbf{R}_{i,i+1} = \mathbf{R}_{i+1,i} = \frac{h_i}{6} \text{ for } i = 2, \dots, n-2 \\ \mathbf{R}_{ij} = 0 \text{ for } |i-j| \geq 2 \end{cases} \tag{3}$$

We have the follwing main theorem

Theorem 1 $\mathbf{R}$ *is positive definite and the pair* $(\mathbf{g}, \boldsymbol{\gamma})$ *specifies a spline function* g *if and only if :*

$$\mathbf{Q}^\top \mathbf{g} = \mathbf{R}\boldsymbol{\gamma},$$

and in that case :

$$\int_a^b g''(x)^2 dx = \boldsymbol{\gamma}^\top \mathbf{R}\boldsymbol{\gamma} = \mathbf{g}^\top \mathbf{K}\mathbf{g},$$

with $\mathbf{K} = \mathbf{Q}\mathbf{R}^{-1}\mathbf{Q}^\top$.

Questions :

1. Using Theorem 1, write the finite dimensional equivalent problem of (1) and show that the solution is characterized by $(\mathbf{g}, \boldsymbol{\gamma})$ and can be written :

   $$\mathbf{g} = (\mathbf{Id}_N + \lambda\mathbf{K})^{-1}\mathbf{y}, \qquad \boldsymbol{\gamma} = \mathbf{R}^{-1}\mathbf{Q}^\top\mathbf{g}$$

   with $\mathbf{y} = (y_1, \cdots, y_N)^\top$.
2. Denote $g_{-i}(x;\lambda)$ the solution of the problem after removing the i-th observation (x_i, y_i) i.e formally

   $$g_{-i}(.;\lambda) = \arg\min_{g\in\text{Splines}} \sum_{n\neq i} |y_n - g(x_n)|^2 + \lambda \int_a^b g''(x)^2 dx$$

   In practice, we would like to set λ that provides the smallest error on *new unseen data*. Thus, we minimize in λ the cross validation (CV) score S defined by

   $$S(\lambda) = \sum_{n=1}^N (y_n - g_{-n}(x_n;\lambda))^2$$

   Consider λ and i fixed and define $\mathbf{y}^{(i)} \in \mathbb{R}^N$ such that $y_j^{(i)} = y_j$ if $j \neq i$ and $y_i^{(i)} = g_{-i}(x_i, \lambda)$. Show that

   $$\mathbf{g}_{-i} = \mathbf{A}(\lambda)\mathbf{y}^{(i)}$$

   where $\mathbf{A}(\lambda) = (\mathbf{Id}_N + \lambda\mathbf{K})^{-1}$.
3. Deduce the formula for $S(\lambda)$:

   $$S(\lambda) = \sum_{n=1}^N \left(\frac{y_n - g(x_n)}{1 - \mathbf{A}(\lambda)_{nn}}\right)^2,$$

   where g is the Spline solution on the entire data $\{(x_1, y_1), \cdots, (x_N, y_N)\}$.
4. Computing $S(\lambda)$ requires the diagonal elements of $\mathbf{A}(\lambda)$. To reduce the computational cost, a simple idea is to replace the denominators by their average value $\frac{1}{N}\text{trace}(\mathbf{A}(\lambda))$. This is known as the generalized cross validation (GCV) score :

   $$S^{(G)}(\lambda) = \frac{\sum_{n=1}^N (y_n - g(x_n))^2}{\left(1 - \frac{1}{N}\text{trace}(\mathbf{A}(\lambda))\right)^2}$$

   Propose an algorithm to compute $S^{(G)}$ for a grid of hyperparameters $\lambda_1, \ldots, \lambda_M$.
5. In practice, it is rather likely to have several observations $y_{n,1}, \ldots, y_{n,m_n}$ for the same x_n. Justify that the problem (with duplicated x_n) is the following one :

   $$\min_{g\in\text{Splines}} \sum_{n=1}^N \sum_{k=1}^{m_n} |y_{n,k} - g(x_n)|^2 + \lambda \int_a^b g''(x)^2 dx$$

   and is equivalent to the weighted regression problem :

   $$\min_{g\in\text{Splines}} \sum_{n=1}^N w_n |y'_n - g(x_n)|^2 + \lambda \int_a^b g''(x)^2 dx$$

   where w_n and y'_n must be determined.
6. Solve the problem and deduce a formula for the modified $\mathbf{A}_w(\lambda)$.